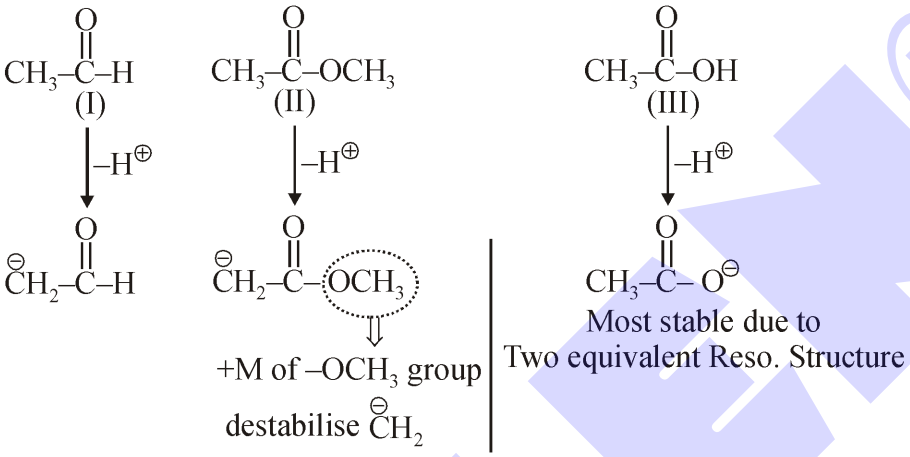
SECTION–I : (i) Only One option correct Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

1. (D)

Sol. Acidic strength ∝ stability of anion



So stability order of anion : – $CH_3-\overset{O}{\overset{\|}{C}}-O^{\ominus}$ > $\overset{\ominus}{C}H_2-\overset{O}{\overset{\|}{C}}-H$ > $\overset{\ominus}{C}H_2-\overset{O}{\overset{\|}{C}}-OCH_3$

and order of acidic strength is ⇒ III > I > II

2. (D)

Sol. $CH_3-\underset{CH_3}{\underset{|}{CH}}-CH_2-CH_3 \xrightarrow{\text{Monochlorination}}$ $CH_2(Cl)-\overset{*}{CH}(CH_3)-CH_2-CH_3$ + $CH_3-C(Cl)(CH_3)-CH_2-CH_3$ (Two Pdts)

$CH_3-CH(CH_3)-\overset{*}{CH}(Cl)-CH_3$ + $CH_3-CH(CH_3)-CH_2-CH_2(Cl)$ (Two Pdts)

* Total monochlorinated products : – 6

(Including stereo.)

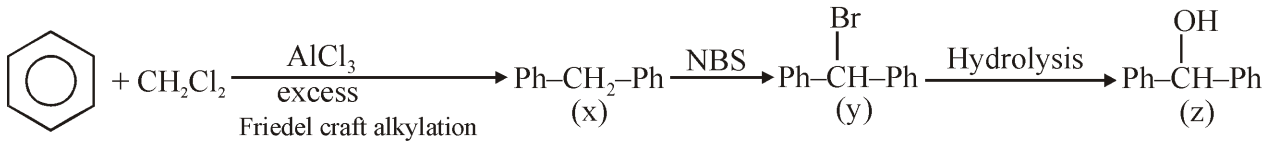
* Total number of fractions = Total Pdts – Enantion Pair

⇒ 6 – 2

⇒ 4

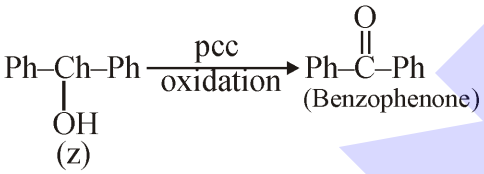
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3. (C)

Sol. 

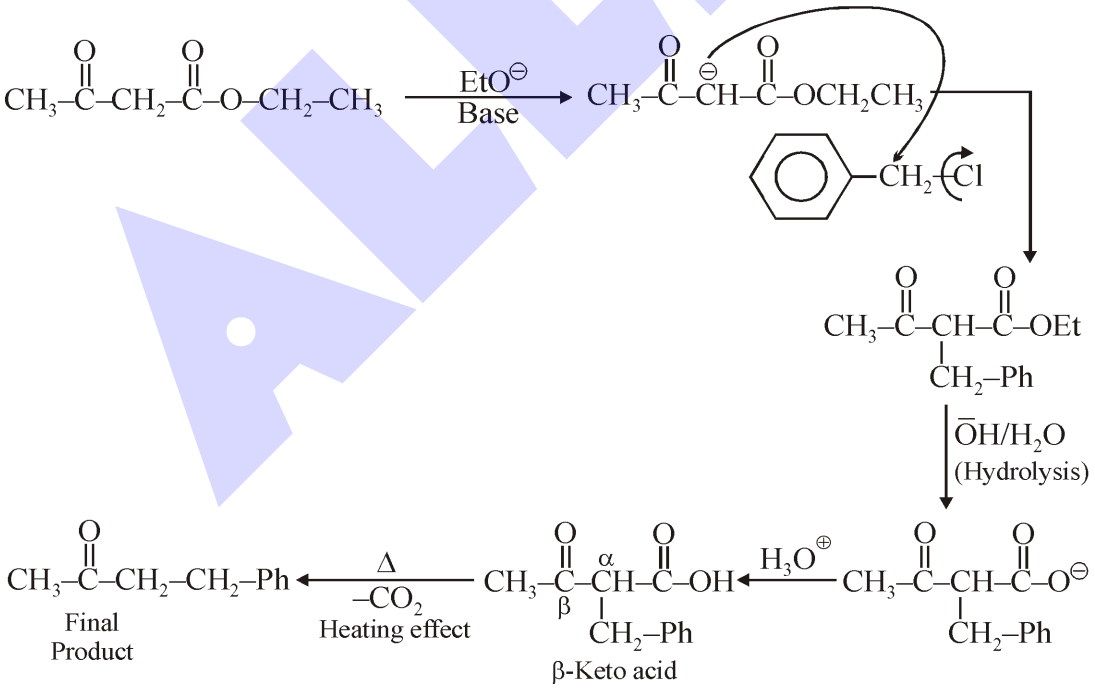
$\Rightarrow$ (A) is incorrect because 'z' gives immediate turbidity with Lucas reagent due to more stable carbocation $(Ph - \overset{\oplus}{C}H - Ph)$

$\Rightarrow$ (B) is incorrect because four benzenoid resonating structures are present in (x)

$\Rightarrow$ (C) is correct statement 

$\Rightarrow$ (D) is incorrect because it gives pale yellow ppt of AgBr$\downarrow$

4. (C)

Sol. 

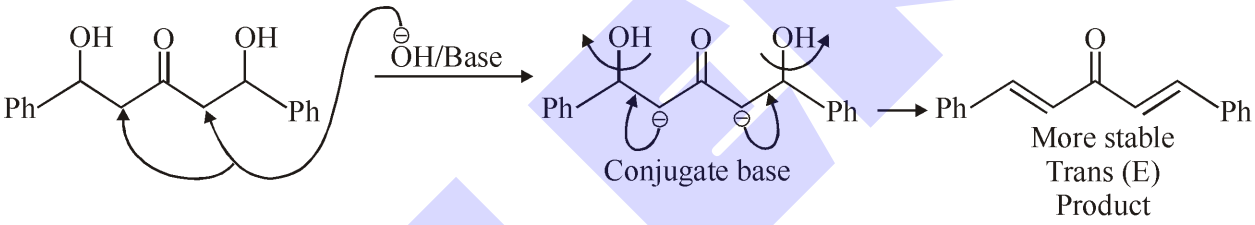
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5. **(C)**

6. **(A,B,D)**

Sol. Et–O–Et
- Na → No Reaction
- $NaNH_2$ → No Reaction
- Aqueous HCl ⇓ Acidic hydrolysis → EtOH + Et – OH
- Aqueous NaOH ⇓ Basic hydrolysis → No Reaction

7. **(A,B,C)**

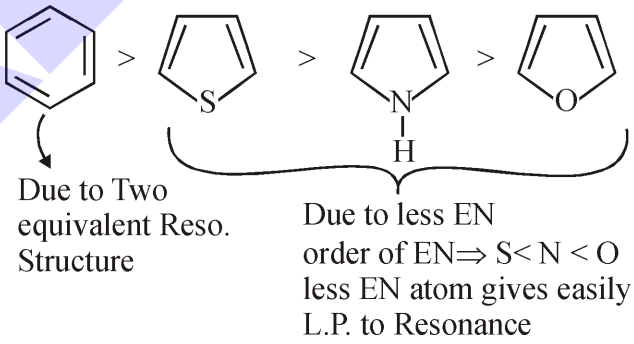
Sol. 

8. **(A,B,C)**

Sol. Informative

9. **(A,B,C)**

Sol. (A) Order of Reso. Energy :-



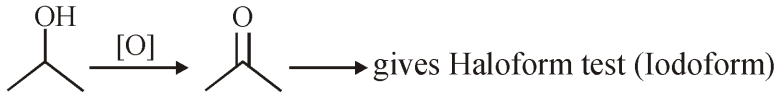
(B) Cyclic form of Glucose is more stable than open chain structure and β-glucopyranose is more stable than α because all (–OH) bonds are equitorial in β-Glucose.

(C) Aromatic Nucleophilic substitution ∝ presence of EWG at ortho & Para position

(D) $\dot{C}F_3$ have more % S character than $\dot{C}Cl_3$ due to high EN

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10. (A,B,D)

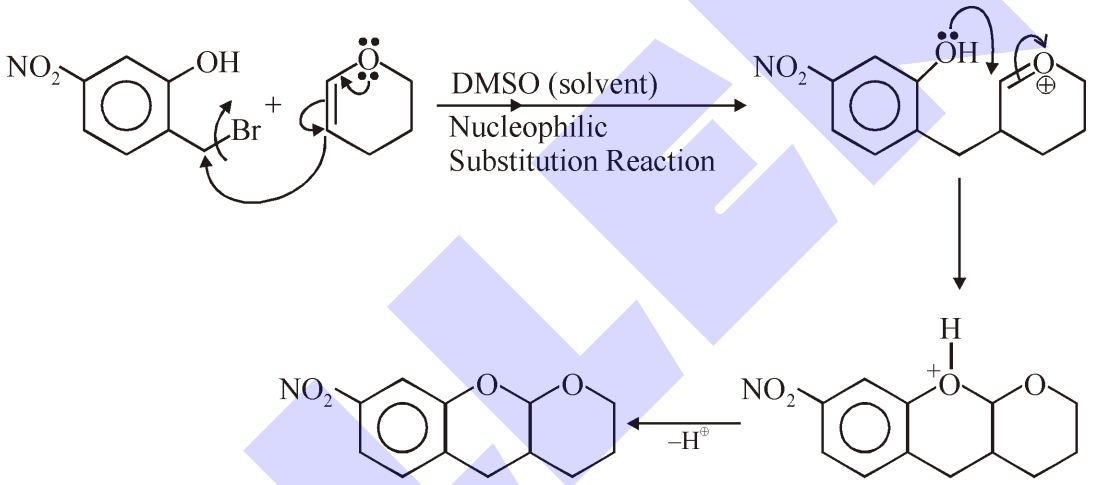
Sol. (A) 

(B) $CH_3-\overset{O}{\overset{\|}{C}}-CH_3$ Gives Iodoform Test

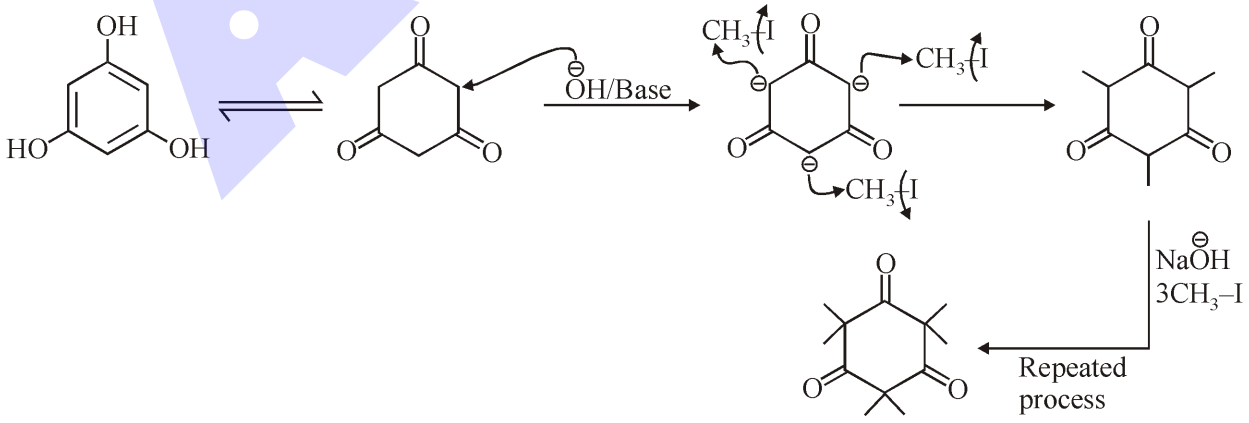
(C) Ester does not give haloform test

(D) $CH_3-\underset{OH}{\underset{|}{CH}}-CH_2-\underset{OH}{\underset{|}{CH}}-CH_3$ gives Iodoform test

11. (A,B)

Sol. 

12. (ABCD)

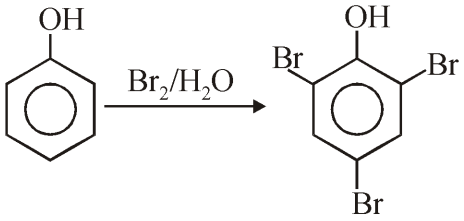
Sol. 

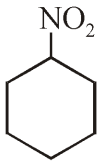
13. (A)

Sol. (P) The Beilstein test is a simple qualitative test for organic halides. ($CH_3CH_2CH_2$–Cl gives this test)

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(Q) $R-\overset{\overset{O}{\|}}{C}-CH_3$ gives Iodoform test due to $-\overset{\overset{O}{\|}}{C}-CH_3$ group

(R) 

(S) 

gives Baker-mulliken test (Test of $-NO_2$ group)

Mulliken-Barker Test–

$$Ph-NO_2 \xrightarrow[NH_4Cl]{Zn} Ph-NH-OH \xrightarrow[\text{Tollen' Reagent}]{2[Ag(NH_3)_2]OH} Ph-N=O + 2H_2O + 4NH_3 + 2Ag\downarrow(ppt)$$

Integer type

1.

Sol. Given → Molar mass = 100

and compounds give 2,4-DNP Test

i.e. compounds are $\underset{(C_nH_{2n}O)}{\text{aldehyde or ketone}}$

$\Rightarrow 12n + 2n + 16 = 100$

$\Rightarrow \boxed{n = 6}$

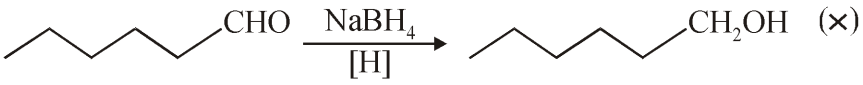
* After reduction with $NaBH_4$, the product alcohol must contain single chiral carbon atom.

* Concept of Resolution →

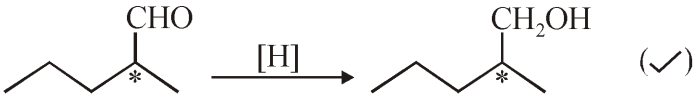
Recemic mixture of 2-methylbutanoic acid (±) + Optically pure isomer of alcohol ⟶ Produce diasteriomeric mixture of ester, which can be separate by fractional distillation

Aldehydes

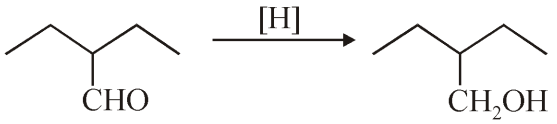
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(I) 

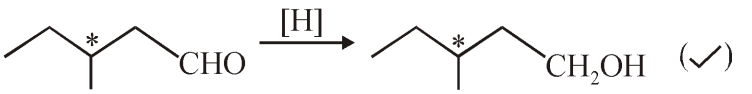
(×)

(II) 

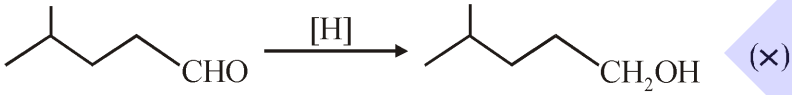
(✓)

(III) 

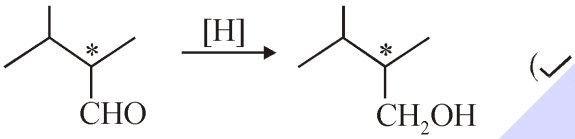
(×)

(IV) 

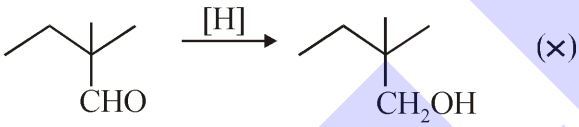
(✓)

(V) 

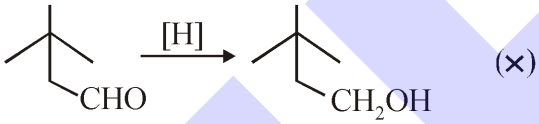
(×)

(VI) 

(✓)

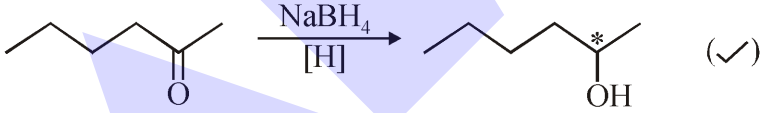
(VII) 

(×)

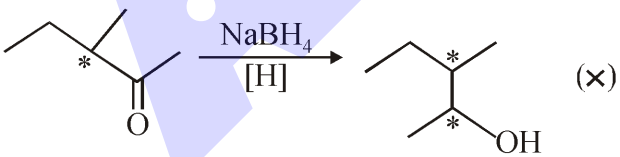
(VIII) 

(×)

Ketones

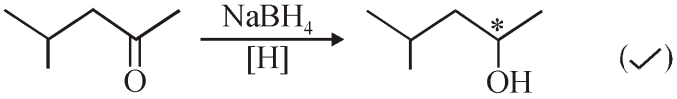
(i) 

(✓)

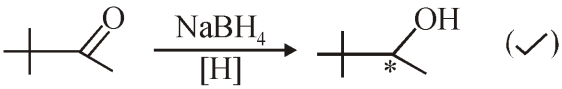
(ii) 

(×)

(2 Chiral carbon)

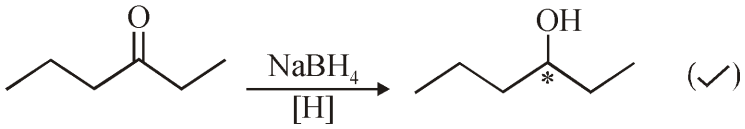
(iii) 

(✓)

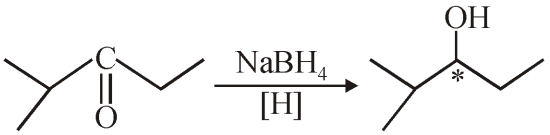
(iv) 

(✓)

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(v) 

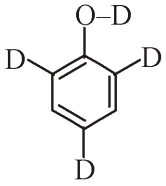
(✓)

(vi) 

(✓)

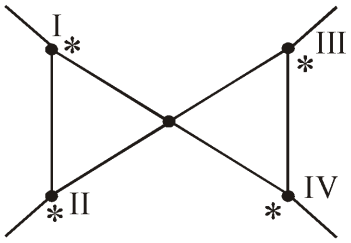
Total (aldehydes + ketone) = 3 + 5 = 8

2. (4)

Sol. 

These 4 hydrogens can be exchanged with D.

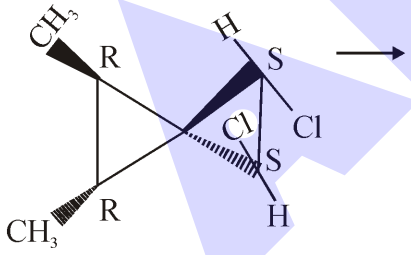
3. (5)

Sol. 

- Optically active isomer = 6

	I	II	III	IV	
(I)	R	R	R	R	] EP
(II)	S	S	S	S	
(III)	R	R	R	S	] EP
(IV)	S	S	S	R	
(V)	R	S	R	S	] EP
(VI)	S	R	S	R	

- Optically inactive isomer = 1



R R S S

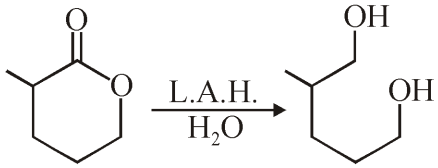
↓

Optically inactive
(having AAOS)

So optically active – inactive = 6 – 1 = 5

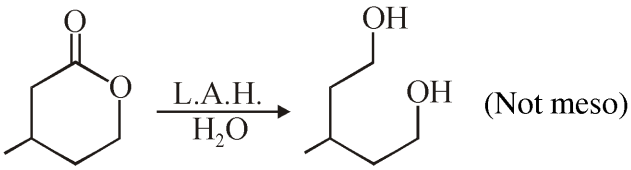
4. (Zero)

Sol. Possible lactones in $C_6H_{10}O$ are :

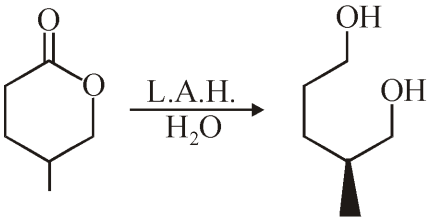


(Not meso)

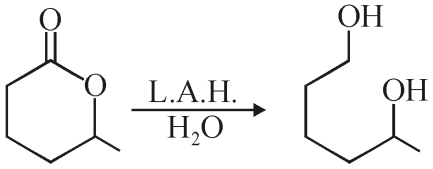
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(Not meso)



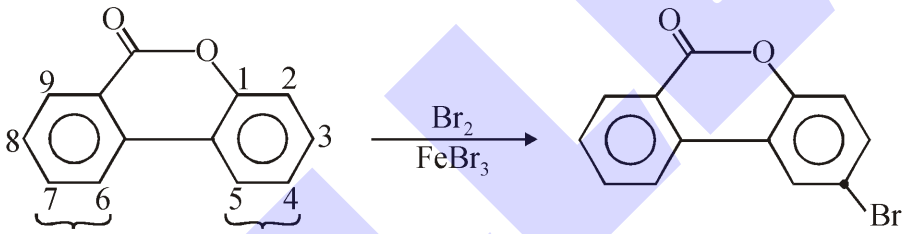
(Not meso)



(Not meso)

5. (4)

Sol.



$\bar{e}$ - deficient benzene ring due to –M effect of $-\overset{\overset{O}{\|}}{C}-$ group.

$\bar{e}$ - rich benzene ring due to –M effect of –O– group

⇓

So this Benzene ring react with electrophile ($Br^{\oplus}$)

→ Due to less crowding
→ Para product (Major)

Section -IV (Matrix-Match Type)

1.

Sol. (A) $CH_3–Cl + \overset{\ominus}{N}aOH \xrightarrow[PAS]{SN^2} CH_3–OH + NaCl$

[Q, R]

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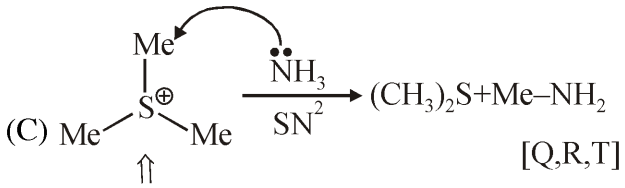
(B) $CH_3–Cl + \ddot{N}H_3 \xrightarrow{SN^2} [H_3\overset{\oplus}{N}\text{------}\overset{\delta+}{C}H_3\text{------}\overset{\delta-}{Cl}]$

[Here Nucleophile (NH_3) is neutral, So in transition state, charge is present, hence polar protic solvent will affect the rate of reaction]

$\downarrow$

$CH_3–NH_2+HCl$

[P,R,T]

(C) 

[Q,R,T]

⇑ PPS will solvate this cation and decrease the rate of reaction, hence PAS will increase rate of reaction

(D) Since product is enantiomeric pair, hence SN^1 mechanism, carbocation is formed as an intermediate and PPS will favour the reaction. [P,S,T]

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